Incident Response Report – VSFTPD Exploitation

Executive Summary

A simulated attack was conducted using Metasploit against a vulnerable FTP service (vsftpd 2.3.4 backdoor) on a Metasploitable2 system. The attack resulted in unauthorized shell access. Wazuh successfully detected the activity, triggering alerts mapped to MITRE ATT&CK technique T1190 (Initial Access). Immediate containment actions were performed, including isolating the affected system and blocking the attacker's IP using CrowdSec. No further lateral movement was observed.

Timeline

Timestamp	Source IP	Alert Description	MITRE Technique
2025-08-18 11:00:00	10.0.2.12	VSFTPD exploit	T1190
2025-08-18 11:02:00	10.0.2.12	Reverse shell detected	T1059
2025-08-18 11:05:00	SOC Action	VM isolated	Containment
2025-08-18 11:07:00	SOC Action	IP blocked (CrowdSec)	Mitigation

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Detection & Triage

- Alerts generated by Wazuh indicated exploitation attempts on port 21
- Log analysis confirmed abnormal FTP service behavior
- Reverse shell activity validated compromise
- Classified as **True Positive – Critical severity**

Response Actions

- Isolated compromised VM from network
- Blocked attacker IP using CrowdSec
- Verified containment:

ping 192.168.1.102

Request timed out (Blocked successfully)

- Initiated forensic data collection

Recommendations

- Patch vulnerable services (remove vsftpd 2.3.4)
- Disable unused services (FTP if not required)
- Implement network segmentation
- Enable IDS/IPS rules for exploit detection
- Conduct regular vulnerability assessments

Stakeholder Briefing

A controlled cybersecurity test identified a vulnerability in a system's file transfer service that allowed unauthorized access. Our monitoring tools quickly detected the activity, and the security team responded immediately by isolating the affected system and blocking the source of the attack. No data loss or spread to other systems was observed. This incident highlights the importance of timely patching and continuous monitoring. Moving forward, we will strengthen system security by updating vulnerable software, improving network controls, and conducting regular security assessments to prevent similar incidents.